

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1504

Tahkenitch 100 kV
July 26, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1504. The Interconnection Customer proposes a 99.8 MW solar photovoltaic facility interconnecting at the Tahkenitch 100 kV substation in the Illinois area. The request was received on June 27, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 545 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,314,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Amax Mines-Tahkenitch 100 kV	Grand Ridge Battery Projects-Gardner 230 kV	124.3	126	56.7
Amax Mines-Gas 100 kV line	Grand Ridge Battery Projects-Gardner 230 kV Line	119.8	120	56.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tahkenitch-Amax Mines Substation 100 kV	Rebuild line with higher-capacity conductor	\$6,586,000	100% (\$6,586,000)
Amax Mines-Gas 100 kV line	Rebuild line with higher-capacity conductor	\$8,728,000	100% (\$8,728,000)
	Total allocated to project		\$15,314,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.